

Justin Proulx

Software Developer • English/French

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Work Experience

Royal Apps – Independent Developer – *Full Time January 2024 to Present, Part Time 2014-2024*

- Designed, developed, released and marketed at least **15 apps** for iOS, Android, Windows, macOS and Linux, using UIKit (**Objective-C**, **Swift**) and Unity (**C#**, **.NET**)
- Analyzed class headers to develop **7 system extensions** in Objective-C which add features to iOS at runtime
- Created numerous web pages using **HTML5**, **CSS** and **JavaScript**
- Over **1.3 million product downloads** and over **\$150,000 in sales**
- Monetized apps using in-app purchases, subscriptions and third-party ads
- Conducted several public beta testing programs through **TestFlight**

Mitel Networks Corp. – Product Development Intern – *May 2022 to August 2022*

- Developed a Jasmine unit testing suite in TypeScript for an Angular app, **improving code coverage by 60%**
- Discovered several bugs in the app, which were reported to the development team
- Presented my findings and unit test designs to the engineering team

ITManager.net – Software Developer, iOS – *May 2021 to August 2021*

- Designed and implemented a new dashboard UI using UIKit and Swift, allowing users to monitor their remote device information and metrics
- Collaborated with Android and web developers to ensure feature parity across versions
- Identified and resolved several bugs in the iOS client, and implemented backlog items

Macadamian Technologies – Software Developer (Intern) – *September 2020 to December 2020*

- Designed and implemented a new admin dashboard in **React** for HealthQ, including reusable widgets with chart rendering support and data processing to ensure chart legibility
- Implemented a “Near Me” screen for the HealthQ React app which uses user location to identify and suggest nearby branches of their organization
- Improved global time zone support in HealthQ by ensuring that local times were correctly displayed in emails and on the web, and correcting the order of the drop-down menu
- Fixed bugs from the backlog and bugs that I discovered, particularly UI bugs and rendering errors

Mitel Networks Corp. – Product Development Intern – *January 2020 to April 2020*

- Researched the possibility of using HTML Custom Elements to allow Mitel’s **Angular** components to work on non-Angular webpages, produced a report and presented my results
- Built several small Angular apps to test Custom Elements in various configurations such as different Angular versions, build tools and build scripts
- Implemented a resizable GIF browser in Angular, which uses Giphy’s public API

Mitel Networks Corp. – Product Development Intern – *May 2019 to August 2019*

- Developed several end-to-end test suites for CloudLink Chat in **TypeScript** using the **Protractor** framework, **reducing the duration of a full test run by 50%**
- Implemented a panel for developers to test different settings configurations in CloudLink Chat
- Developed an Angular app for testing the CloudLink Chat component, featuring a preference panel, mobile support and a sidebar for managing conversations (with streaming support via Angular Observables)

Education

University of Ottawa – B.A.Sc. Software Engineering (Co-op) – *September 2018 to December 2023*

- Capstone: React client and a **Node.js/Express** server, both built in JavaScript. Server hosted on Azure, with GitHub Actions for deployment and Cosmos DB for the database
- Elected Software Engineering Commissioner for the IEEE student branch
- Appointed webmaster for Chance to EdVance club
- Member of the Google Developer Student Club uOttawa Chapter
- Part of the winning team for the programming category at the uOttawa Engineering Competition in 2021

Skills

Programming Languages & Frameworks

Swift • C# • Objective-C • TypeScript • JavaScript • HTML5 • CSS • Python • React • Angular • Node.js

Tools

Git • VS Code • Xcode • Unity • Slack • Notion • Kanban • MS Office • Adobe CC (PS, AE) • Logic Pro

Highlighted Projects

Impeached 2 – iOS, Android, Windows, macOS, Linux – *Initial Release July 2023*

- Sequel to my most downloaded App Store app, *Impeached: Be The President* (released 2018)
- Unity project implemented in C#
- **4.5-star average rating** on the App Store (both U.S. and global, with over 1000 total ratings) as of June 2026
- Includes original text content, pixel art and music
- Features a Deck Editor for creating and sharing custom content, and supports adding new endings and graphical overlays to the game
- Allows sharing user generated content via a steganographic image, which is resistant to social media platform JPEG compression
- Multiplatform release via the App Store, Google Play and Steam

Bankrupt: Corporate Warfare – iOS, macOS – *Initial Release May 2026*

- 3D real-time strategy game with economic elements
- Implemented in Swift, using UIKit and RealityKit
- Multithreaded using **Swift Concurrency** (actors and async/await pattern)
- Optimized to run on older devices using techniques like chunking and mesh instancing, driven by data obtained from diagnostic tools like the Xcode profiler and Instruments
- Modular greedy algorithm-based AI system to control computer players, for which new behaviors can easily be added
- Underwent several major rewrites during development, including upgrading the graphics from flat 2D to isometric 2D, and later to 3D, and switching from turn-based gameplay to real-time
- Includes single-player campaign and a Level Editor to allow for player-designed maps with custom mission objectives
- Supply and demand market system to dictate the value of in-game resources
- Basic fluid simulation to allow for more realistic lake placement, and allow for terrain flooding
- Procedurally generated worlds which can have a variety of planetary compositions, which affects the availability of different resources and technologies (and thus, the market value for those resources)
- Includes original pixel art textures, music, level designs and text

ProKeys – iOS – *Initial Release April 2021*

- Successor to my popular iOS system extension, *RealKeys* (released 2020)
- Implemented mainly in Objective-C, with some Objective-C++
- Added a multitude of customization features to the stock iOS keyboard, including expanded haptic feedback options, custom colors and keystroke animations
- Noted at time of release by the store maintainers as having “**some of the best marketing style on the platform**”
- Required independent research into the undocumented classes that power the iOS default keyboard, and I have since been able to provide support to other keyboard extension developers who were running into the same issues

Jellyfish – iOS – *Initial Release February 2019*

- Implemented mainly in Objective-C, with some Objective-C++
- Redesigned the iOS lock screen, particularly the date & time element, and added many customization options otherwise not available to iOS users. Also added a weather widget, a feature not available at the time of Jellyfish’s release
- Required research into undocumented classes that make up the iOS lock screen and those responsible for acquiring weather data
- Ran a successful marketing campaign on Twitter, where I collaborated with larger accounts to run giveaways of Jellyfish. This campaign was responsible for a **4x increase** in my account’s follower count

Mind Dash – iOS, macOS – *Initial Release March 2025*

- Successor to my previous math game *Speed Math* (released 2017)
- Implemented in Swift, using UIKit
- Math-based puzzle game with procedurally generated daily puzzles
- Daily puzzles are generated offline, yet still synchronized for all players worldwide
- Custom math expression generation logic ensures that all expressions are valid and that all puzzles are solvable
- Includes online leaderboards via Game Center, and a score card sharing system that allows players to challenge their friends using just an image or a URL